Promotion:

In the promotion phase, the initiated cells multiply and a new abnormal tissue with no useful function is formed. This is called **"neoplasm".** Promotion takes place by a continuous exposure of the initiated cells to promoting agents.

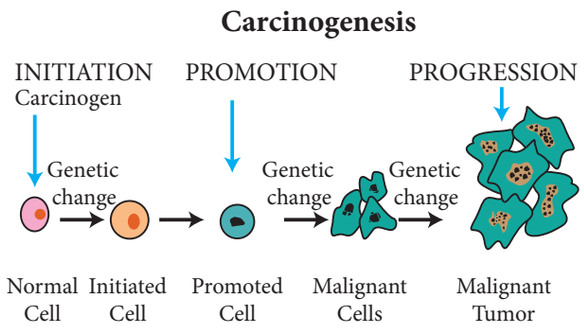


Fig 12.2 Genetic mutation of a normal cell to a cancer cell

Progression:

During the progression phase, the abnormal tissue grows and leads to the development of a fully malignant tumor. This tumor has the capacity to invade normal tissues which may ultimately lead to the spread of cancer to other parts of the body. This is called as **"metastasis."** A Cancer that has spread to other parts of the body is said to have "metastasized".

A tumor is an abnormal growth of cells that serves no purpose.

There are two types of tumors : Malignant and benign tumors

A benign tumor is a non-cancerous tumor that does not spread to other parts of the body. A malignant tumor is a cancerous tumor that may spread to other parts of the body

12.2 Classification of cancer

Cancers are of four main types based on the type of cell that produces it. They are carcinoma, sarcoma, lymphoma, and leukemia.

1. **Carcinoma** is cancer of the epithelial cells that line the digestive tract, urinary bladder and the uterus.
2. **Sarcoma** is cancer of the soft tissues of the body, such as muscle, fat, nerves, tendons, blood and lymph vessels.
3. **Lymphoma** is cancer of the lymphoid tissue. This includes the lymph nodes, bone marrow, spleen, and thymus gland.
4. **Leukemia** develops from the white blood cells and also affects the bone marrow and spleen.

The most common cancers in men are that of the lung, colon, pancreas, liver and prostrate gland. In women, cancer of the breast, ovary, cervix, uterus and colon are common. Besides this, cancer can occur in any part of the body like the thyroid gland, stomach, oral cavity, esophagus, skin, brain etc.

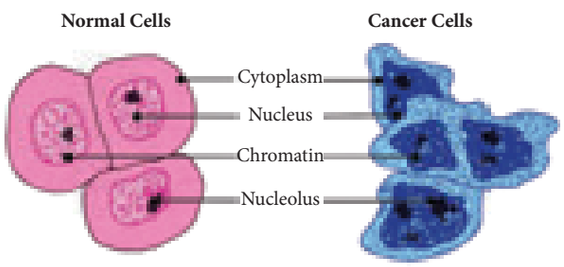


Fig 12.3 Comparison between a normal cell and a cancer cell

Carcinogenesis is the process in which normal cells turn into cancer cells.

Alterations (mutations) in DNA/ genes occur during carcinogenesis. A carcinogen is a physical, chemical or viral agent that induces cancer by bringing about changes in the DNA.

UNIT - 12.indd 192 25-02-2019 15:39:01